

VR-Sets. Preprint.

Part IV. The Two Modes: ZFC and ZFA

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IV.1. The fork: two modes as two levels of application of the closure principle

In Part III the closure principle was applied to describable operational functionalities without restriction: every describable functionality is a set. In §III.8 it emerged that one of the ZF axioms — foundation — does not hold on all such functionalities. The Quine atom $A = \{A\}$ showed that the description "functionality revealing itself" is operationally describable but gives rise to a set with an infinitely descending chain of membership.

This is the ontological fork of VR-Sets. The closure principle in full strength gives a universe in which all ZF axioms hold **except foundation**. The closure principle, restricted by the condition that successive unfoldings terminate at $\emptyset$, gives a universe in which **all** ZFC axioms hold, including foundation. These two universes are the two modes of VR-Sets:

- **ZFC-mode**: the closure principle is applied only to functionalities whose successive unfoldings terminate at $\emptyset$.
- **ZFA-mode**: the closure principle is applied to all describable functionalities, without additional restriction.

In this part the modes are given a precise definition; it is proved that all ZFC axioms hold in the ZFC-mode (an assembly of the results of Part III); and the conjecture is discussed that the ZFC-mode is equivalent to classical ZFC and the ZFA-mode to the theory of Aczel (AFA, 1988). The corresponding converse directions — that every set of classical ZFC has an operational representation in the ZFC-mode of VR-Sets, and analogously for AFA — remain open; for each of them §IV.5 and §IV.6 analyse what is required for a proof.

Remark: the modes are not temporal

In the ontology of VR-Sets only $\emptyset$ is ontologically primary; operability is derived (not a self-standing kind of being, but action upon $\emptyset$ and upon the results of prior actions; see Part I §I.3 in detail). Considered as a description, however, operability is an integral object, not a process in time. When a functionality is given by its description, it **is** that description fully — there is no "moment of construction," no "already-built" or "being-built-just-now" in the ontology. Queries to a functionality are a way of addressing it, not moments of its being.

The distinction between the ZFC-mode and the ZFA-mode is therefore **not temporal** ("did the set manage to be built before reference was made to it?") but **structural** — a property of the functionality itself, considered as an integral object. The property is whether successive unfoldings of its responses terminate at $\emptyset$ in a finite number of steps.

This distinguishes the formulation of VR-Sets from the standard set-theoretic approach, where the well-foundedness condition is often described through an "iterative conception of the set" — the image of stepwise construction. In VR-Sets there are no steps and no construction; there is a structural property of a functionality.

IV.2. Unfolding, termination, operational depth

Before the formal definition of the modes we sharpen the technical apparatus inherited from Part II.

Unfolding of a functionality

Definition IV.1 (successive unfolding). Let A be an operational set. A *successive unfolding* of the functionality A is any sequence of sets

$$A = x_0 \ni x_1 \ni x_2 \ni x_3 \ni \dots$$

where x_{k+1} is one of the elements revealed by the functionality x_k . The unfolding *terminates* at $\emptyset$ if there exists n such that $x_n \equiv \emptyset$. The unfolding *does not terminate* if no such n exists.

The property of termination

Definition IV.2 (well-founded functionality). A functionality A is *well-founded* if **every** successive unfolding starting with A terminates at $\emptyset$ in a finite number of steps.

An equivalent formulation: there is no infinite sequence $x_0 \ni x_1 \ni x_2 \ni \dots$ starting with A . This is precisely the classical well-foundedness of the membership relation, lifted to an individual functionality.

Relation to operational depth

Lemma 3 of Part II defined the *operational depth* of a set as the least number of unfolding steps after which $\emptyset$ is reached. Cyclic sets (such as the Quine atom) have no finite depth because unfolding does not terminate.

Assertion IV.1. A functionality A is well-founded if and only if every set reachable from A by a finite chain of membership has a defined operational depth (finite, or ω -depth without cycling).

This is not a theorem in the strict sense but a development of definitions: well-foundedness is precisely the absence of cycles and the absence of infinite descending chains. For us it is important to fix that well-foundedness is a **structural property** of a functionality, verified on its description, not acquired at a moment of construction.

Examples

- $\emptyset$ is well-founded trivially (no elements, no unfoldings).
- Every VR number (O_n) is well-founded: its depth is n , and the single unfolding chain terminates at $\emptyset$ in n steps.

- ω is well-founded: any unfolding starting with ω yields on the first step some O_n with finite n , and then terminates in n steps. Despite the "infinity" of ω itself, each concrete unfolding is finite.
- The Quine atom $A = \{A\}$ is **not** well-founded: the unfolding $A \ni A \ni A \ni \dots$ does not terminate.
- The set $B = \{\emptyset, B\}$ (containing $\emptyset$ and itself) is not well-founded: the chain $B \ni B \ni B \ni \dots$ does not terminate, even though the chain $B \ni \emptyset$ does terminate.

The last example shows that well-foundedness is a property of **all** unfoldings, not of the existence of at least one terminating chain. This is in accord with the classical definition.

IV.3. Formal definitions of the ZFC-mode and the ZFA-mode

Having the notion of well-foundedness as a structural property of a functionality, we define the modes.

Definition IV.3 (ZFC-mode)

The **ZFC-mode** of VR-Sets is the operational universe in which the closure principle is applied only to well-founded operational functionalities. Formally: a set exists in the ZFC-mode if and only if (a) it is operationally describable and (b) it is well-founded.

Definition IV.4 (ZFA-mode)

The **ZFA-mode** of VR-Sets is the operational universe in which the closure principle is applied to all describable operational functionalities, without the well-foundedness restriction. Formally: a set exists in the ZFA-mode if and only if it is operationally describable.

Relation between the modes

Assertion IV.2. Every set of the ZFC-mode is a set of the ZFA-mode: $\text{ZFC-mode} \subseteq \text{ZFA-mode}$ as operational universes. The inclusion is strict: the Quine atom $A = \{A\}$ belongs to the ZFA-mode but not to the ZFC-mode.

The proof is trivial: the ZFC-mode is defined as the ZFA-mode with the additional condition of well-foundedness, so $\text{ZFC} \subseteq \text{ZFA}$. Strictness of the inclusion is witnessed by the Quine atom — it is describable (the functionality "reveal yourself" gives a definite response to every query) and not well-founded (the chain $A \ni A \ni \dots$ does not terminate).

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Remark: the modes are not paradoxical

The Quine atom in the ZFA-mode is not paradoxical. Russell's paradox (Part II §5.2) is dissolved by the closure principle: the description $R = \text{"reveal } x \text{ if and only if } x \notin x\text{"}$ requires a contradictory response to the query " $R \in R?$ " and so does not specify a describable functionality. The Quine atom carries no such contradiction: "the functionality A reveals A " is a definite description, specifying a definite response to every query. That A **contains itself as an element** does not contradict the ontology of VR-Sets, because membership in VR-Sets

is reference, not physical containment (Foundation Notes §2). $A \in A$ is "the functionality A in one of its responses returns A ," which is operationally defined.

The distinction between the ZFC-mode and the ZFA-mode is therefore not a distinction between "non-contradictory" and "paradoxical." Both modes are non-contradictory. The distinction lies in which operationalities are admitted.

IV.4. Theorem: all ZFC axioms hold in the ZFC-mode

This section assembles the results of Part III as they apply to the ZFC-mode and establishes a direct inclusion: the ZFC-mode of VR-Sets is a model of classical set theory ZFC.

Theorem IV.1 (completeness of ZFC in the ZFC-mode)

In the ZFC-mode of VR-Sets all ZFC axioms hold: extensionality, empty set, pairing, union, power, infinity, replacement, foundation, choice.

Proof

We show that for every ZFC axiom the corresponding closure theorem of Part III remains in force under restriction to well-founded sets — that is, that the constructions of Part III do not lead outside the ZFC-mode.

Extensionality (§III.1). Lemma 1 of Part II is formulated through $\equiv$ and applies to any sets, in particular to well-founded ones. The well-foundedness condition does not affect the condition of coincidence of functionalities.

Empty set (§III.2). $\emptyset$ is well-founded trivially (no elements means no unfoldings). $\emptyset$ belongs to the ZFC-mode.

Pairing (§III.3). Let a and b be well-founded. The functionality $\{a, b\}$ reveals a and b . Any unfolding starting with $\{a, b\}$ yields on the first step a or b — a well-founded set — and then the chain terminates by assumption. Hence $\{a, b\}$ is well-founded. The ZFC-mode is closed under pairing.

Union (§III.4). Let A be well-founded. $\cup A$ reveals elements of elements of A . Any unfolding of $\cup A$ yields on the first step some $x \in C$ for $C \in A$; the depth of x is strictly less than the depth of A (two steps down membership); by the well-foundedness of A the chain terminates. The ZFC-mode is closed under union.

Power (§III.5). Let A be well-founded. $\wp(A)$ reveals all subsets of A — operationally describable functionalities revealing subsets of the elements of A . Each subset of A is well-founded (its elements are elements of A , well-founded by assumption; an unfolding one step down lands on a well-founded element of A). $\wp(A)$ is itself well-founded: its unfolding on the first step yields a well-founded subset of A . The ZFC-mode is closed under the power operation.

Infinity (§III.6). ω is well-founded: this was shown in §IV.2 (any unfolding of ω on the first step yields O_n with finite n , and the chain terminates in n steps). ω belongs to the ZFC-mode.

Replacement (§III.7). Let A be well-founded and F a describable procedure mapping well-founded sets to well-founded sets. Then $\text{image}(F, A)$ consists of well-founded sets; the chain of unfolding $\text{image}(F, A)$ on the first step yields $F(x)$ — well-founded — and terminates by assumption. The ZFC-mode is closed under describable transformations preserving well-foundedness.

Foundation (§III.8). This is the defining property of the ZFC-mode: every set of the ZFC-mode is well-founded by Definition IV.3. Foundation holds automatically.

Choice (§III.9). Let $\{A_i\}_{i \in \omega}$ be a countable family of non-empty well-founded sets. The choice functionality reveals the first element of each A_i . The elements are well-founded (elements of well-founded sets are well-founded). The chain of unfolding choice on the first step yields a well-founded element — and terminates. The ZFC-mode is closed under countable choice, which on a countable operational universe yields full AC.

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Corollary

Corollary IV.1. The ZFC-mode of VR-Sets is a model of the theory ZFC: every theorem of ZFC expressed in the language of operational sets holds in the ZFC-mode.

This is a direct consequence of Theorem IV.1 and of the fact that ZFC is precisely the theory derivable from these axioms. Substantively: the whole of standard mathematics formalisable in ZFC is accessible in the ZFC-mode of VR-Sets.

From this, however, it does **not** follow that the ZFC-mode is equivalent to ZFC. Equivalence requires the converse direction: that every model of ZFC is realised in the ZFC-mode of VR-Sets. This is an open question, treated in §IV.5.

IV.5. Conjecture: the ZFC-mode is equivalent to classical ZFC

The direct direction has been established: the ZFC-mode is a model of ZFC. The converse direction — that every model of ZFC is realised in the ZFC-mode of VR-Sets — is formulated as a conjecture, and we examine what is required for its proof.

Statement

Conjecture IV.1. The ZFC-mode of VR-Sets and classical ZFC are mutually interpretable: there exist operational interpretations of the standard model of ZFC in the ZFC-mode and conversely, agreeing on the membership relation.

What is already proved

The direct direction $\text{ZFC-mode} \rightarrow \text{ZFC}$ follows from Theorem IV.1: every statement provable in ZFC holds in the ZFC-mode, since all axioms hold. Accordingly, any operational construction correct in the ZFC-mode has an analogue in ZFC.

What is required for the converse direction

The converse direction $ZFC \rightarrow ZFC\text{-mode}$ requires that for every set of classical ZFC there exist an operationally describable representation in the ZFC-mode. This is stronger than "a model exists": classical ZFC proves the existence of sets for which an operational description is not obvious.

Principal obstacle: the uncountability of $\wp(\omega)$ in classical ZFC. In classical ZFC, $\wp(\omega)$ is uncountable (Cantor's theorem). In the ZFC-mode of VR-Sets, $\wp(\omega)$, as shown in §III.5, is countable: there are countably many operationally describable subsets of ω . This means that the classical $\wp(\omega)$ is **not** realised in the ZFC-mode as a single object.

Possible responses to this obstacle:

(a) *Conjecture in a weakened form.* To assert mutual interpretability not with respect to full classical ZFC but with respect to the *constructive part* of ZFC — the fragment yielding only operationally describable sets. This formulation is correct and provable, but less ambitious: it asserts the correspondence only for "constructive" mathematics.

(b) *Conjecture via submodels.* The ZFC-mode is interpreted as a countable transitive model of ZFC (or, more precisely, a fragment of one). By the Löwenheim–Skolem theorem such a model exists. The correspondence then is not between the ZFC-mode and the "real" classical ZFC, but between the ZFC-mode and one of the countable models of ZFC. This is consistent with the conjecture in the form "there is a countable model of ZFC realised in the ZFC-mode," and this direction appears the most promising for a rigorous proof.

(c) *Conjecture as a declaration of difference.* To acknowledge that the ZFC-mode of VR-Sets and full classical ZFC are **not** equivalent — they diverge precisely on those objects that classical ZFC proves to exist but which VR-Sets cannot specify operationally (such as uncountable subsets of ω). In this formulation the conjecture becomes a theorem on divergence, and the work in §IV.5 becomes its formal fixation.

Our working position is (b): the ZFC-mode of VR-Sets corresponds to a **countable model** of classical ZFC. This position is technically realistic (countable models of ZFC are well studied), preserves the substantive connection with ZFC, and is consistent with the countability of the operational universe fixed in the Foundation Notes. A detailed proof of such a correspondence is a separate technical undertaking, beyond the scope of this preprint.

Methodological remark

The fact that the ZFC-mode of VR-Sets is not the "whole" of classical ZFC but a countable model of it is, in the VR system, not a defect but an expected consequence of the operational ontology. "Only $\emptyset$ is, all else is doing" means that the ontology of VR is restricted to what can be specified operationally; operationally specifiable objects are finitely or countably many. The uncountable objects of classical ZFC are an artefact of the non-operational ontology of classical mathematics, which admits the existence of objects without their operational description.

VR-Sets therefore does not claim to reproduce the full classical universe of sets. It claims to be an ontologically coherent alternative in which all objects are operational and which nonetheless contains enough for the whole of constructive and countable mathematics.

IV.6. The ZFA-mode and Aczel

A parallel picture is built for the ZFA-mode. Aczel (1988) proposed the theory AFA (Anti-Foundation Axiom) — an extension of ZF in which foundation is replaced by anti-foundation: every directed graph is realised as the membership graph of a unique set. This permits the existence of cyclic sets such as the Quine atom.

Theorem IV.2 (completeness of ZFA in the ZFA-mode)

In the ZFA-mode of VR-Sets all ZFC axioms hold **except foundation**, and also the anti-foundation axiom AFA.

Proof

The axioms of extensionality, empty set, pairing, union, power, infinity, replacement, and choice are closure theorems of Part III. The proofs do not use well-foundedness and carry over to the ZFA-mode directly.

Foundation in the ZFA-mode **does not** hold: the Quine atom $A = \{A\}$ belongs to the ZFA-mode (a describable functionality) and violates foundation. This is in accord with the removal of the corresponding condition in Definition IV.4.

AFA in the ZFA-mode. Aczel's anti-foundation axiom asserts: for every directed graph G there exists a unique function assigning to each vertex a set so that the membership relation between these sets agrees with the edge relation of the graph. In the operational formulation: for every describable graph G (a graph whose vertices and edges can be operationally enumerated) there exists an operationally describable collection of functionalities realising the graph.

The describability of such a collection is a standard construction: every vertex v of the graph G specifies a functionality f_v which on a query reveals f_w for every edge $v \rightarrow w$. This is an operationally describable description (for describable G). By the closure principle in the ZFA-mode (without the well-foundedness condition) this is a collection of sets. Uniqueness up to bisimulation is a consequence of extensionality (Lemma 1 of Part II).

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Remark: operational AFA vs. classical AFA

Classical AFA (Aczel 1988) ranges over *every* directed graph, including uncountable graphs without a finite description. The theorem above establishes AFA in the operational ontology only for *describable* graphs — those whose vertices and edges admit a finite operational description. Call this *operational AFA*. This restriction is the exact parallel of the situation with classical ZFC and the ZFC-mode in §IV.5: the operational universe is countable, so its anti-foundation principle ranges only over countable describable graphs. For every classical undescribable graph there is no corresponding object in VR-Sets, just as there is no

uncountable $\wp(\omega)$. The symmetry between the two modes is therefore: in the ZFC-mode the operational universe is a countable analogue of classical ZFC; in the ZFA-mode it is a countable analogue of classical ZFC- + AFA. Conjecture IV.2 below asserts mutual interpretability via a countable model in this same sense.

Conjecture IV.2 (equivalence with AFA)

Conjecture. The ZFA-mode of VR-Sets and the theory ZFC- + AFA are mutually interpretable (in the sense analogous to Conjecture IV.1: presumably via a countable model).

Obstacles and approaches parallel those of §IV.5. The direct direction *ZFA-mode* $\rightarrow$ *AFA* follows from Theorem IV.2. The converse direction faces the same difficulties with uncountability; the working solution is a correspondence with a countable model of AFA.

IV.7. The Quine atom and cyclic sets: examples in the ZFA-mode

To make concrete how the ZFA-mode works in practice, we examine several examples of non-well-founded sets and verify that they are ontologically legitimate in VR-Sets.

Example IV.1: the Quine atom

Description: the functionality A reveals A on every query. Formally: $A = \{A\}$.

Operationality: on a query the functionality gives a response — A . This is a definite response; the functionality is operationally describable. By the closure principle (in the ZFA-mode) A is a set.

Well-foundedness: the chain $A \ni A \ni A \ni \dots$ does not terminate. A does **not** belong to the ZFC-mode.

Uniqueness: by extensionality there is exactly one set with this functionality. By AFA, every graph consisting of a single vertex with a self-loop is realised in a unique way — this is A .

Example IV.2: a pair of mutually referring sets

Description: sets B and C such that $B = \{C\}$ and $C = \{B\}$.

Operationality: the functionality B on a query reveals C . The functionality C on a query reveals B . Both descriptions are definite.

Unfolding chain: $B \ni C \ni B \ni C \ni \dots$ does not terminate. Neither B nor C belongs to the ZFC-mode.

Distinguishability: B and C are different sets (B contains C , C contains B ; the functionalities do not coincide), but they are *bisimilar*: their membership graphs are isomorphic (a two-vertex cycle). In AFA, bisimilar sets are equal; in the ZFA-mode of VR-Sets, extensionality through $\equiv$ gives the same result if $\equiv$ is interpreted as bisimulation. This is consistent with the standard AFA literature.

Example IV.3: a set partially well-founded

Description: $D = \{\emptyset, D\}$. The functionality reveals $\emptyset$ and D itself.

Operationality: on the 1st query D reveals $\emptyset$; on the 2nd — D ; on the 3rd, exhaustion (or D again, depending on the formulation). The description is definite.

Well-foundedness: the chain $D \ni \emptyset$ terminates at the first step. The chain $D \ni D \ni D \ni \dots$ does not terminate. Since **not every** chain terminates, D is not well-founded by Definition IV.2. D belongs to the ZFA-mode but not to the ZFC-mode.

This example shows that well-foundedness demands the termination of **all** chains, not the existence of at least one terminating chain. This is the standard formulation, here verified on concrete operational descriptions.

Example IV.4: an infinitely descending series

Description: a set E_0 such that $E_0 = \{E_1\}$, $E_1 = \{E_2\}$, $E_2 = \{E_3\}$, ..., and this sequence does not terminate.

The description of E_0 operationally requires the definition of all E_n — a countable series of functionalities. This is operationally describable provided there is a uniform description of the series (for instance, an indexed description of E_n by n).

Unfolding chain: $E_0 \ni E_1 \ni E_2 \ni \dots$ does not terminate. E_0 is not well-founded; it belongs to the ZFA-mode but not to the ZFC-mode.

The example shows that non-well-foundedness is not necessarily connected with self-reference: one can construct a descending series without a cycle. The ZFA-mode encompasses both kinds of such sets.

Remark: the ZFA-mode as an ontological option

All these sets are legitimate objects of the ZFA-mode. They carry no paradox (the closure principle has been checked on each description); they create no contradiction with the other axioms (Part III and Theorem IV.2 were proved without the well-foundedness condition).

The question of **which** mathematical applications such sets have is beyond the scope of this preprint. In the standard literature, AFA is applied to the semantics of circular phenomena: processes, recursive data structures, situational semantics (Barwise–Moss 1996), the theory of matchings with cyclic references. For VR-Sets it is important to fix: these applications are **ontologically legitimate**, and the passage from the ZFC-mode to the ZFA-mode is not a departure from rigour but an explicit ontological choice.

IV.8. Relation to standard approaches

We briefly fix how the VR-Sets formulation of the two modes relates to standard approaches in set theory.

Aczel (1988)

Aczel introduced AFA as an anti-axiom: "every pointed graph has a unique realisation as a set." In VR-Sets this axiom is a theorem of the ZFA-mode (see §IV.6). The difference lies in the manner of formulation: in Aczel it is a postulate of existence; in VR-Sets a consequence of the closure principle applied to describable graphs.

Bisimulation in Aczel is the equivalence relation on sets identifying sets with isomorphic membership graphs. In VR-Sets, the operational identity $\equiv$ (Definition 4 of Part II) is structurally the same: coincidence of functionalities by elements, recursively up to $\equiv$. This gives a direct correspondence: $\equiv$ in VR-Sets = bisimulation in Aczel.

Barwise–Moss (1996)

Barwise and Moss in "Vicious Circles" developed in detail the applications of AFA to situational semantics and the theory of non-correspondence (non-well-founded sets for modelling self-reference in natural language). VR-Sets is compatible with these applications: the ZFA-mode gives the same mathematical universe as standard AFA, up to Conjecture IV.2.

The iterative conception of the set

The standard picture of ZFC rests on the iterative conception: sets are built in stages of ordinals ($V_0 = \emptyset$, $V_{\alpha+1} = \wp(V_\alpha)$, with union at limit steps). This picture is *temporal*: a set exists from the stage at which it first appears.

In VR-Sets there is no such temporality (§IV.1). The ZFC-mode is not "built up in stages" but is defined by a structural property — well-foundedness. This is not a different theory; it is a different *manner* of describing the same (or, in view of §IV.5, the operationally accessible part of the) universe. The iterative conception and the operational one are two different methodological optics on a single mathematical structure.

Topoi and categorical set theory

In categorical set theory (Lawvere, ETCS, ETCC) the accent is shifted from "membership" to "functions." VR-Sets is closer to this tradition than to classical ZF: the membership functionality is a category of arrows "reveal an element," and $\equiv$ is an isomorphism in this category. The connection of VR-Sets with topoi is a separate substantive programme, indicated in Part IX as an open question.

Conclusion

In Part IV the fork between the ZFC-mode and the ZFA-mode has been formalised as two levels of application of the closure principle. The principal results:

- (1) Definitions IV.3 and IV.4 formulate the modes through a *structural* property of a functionality (well-foundedness — termination of all successive unfoldings at $\emptyset$), without introducing a temporal order of construction and without a separate layer of labels.
- (2) Theorem IV.1 establishes that the ZFC-mode is a model of classical ZFC: all nine axioms hold.
- (3) Theorem IV.2 establishes an analogous result for the ZFA-mode: all ZFC axioms hold except foundation, and AFA holds.

(4) Conjectures IV.1 and IV.2 formulate equivalence with the classical theories, with an analysis of what is required for the converse directions (the working solution: correspondence via a countable model).

(5) Examples IV.1–IV.4 make concrete the ZFA-mode by means of the Quine atom, cycles, partially well-founded sets, and infinitely descending series.

Methodological summing-up. The two modes are not two different "kinds" of sets and not two competing foundations, but two ontological options of a single system. The closure principle and the definitions of Part II operate in the same way in both modes. The difference is the explicit choice of which operationalities to admit: only the well-founded ones (standard mathematics) or all the describable ones (mathematics with self-references and cycles).

This choice is made **once and explicitly**, not in each particular construction. After the mode is chosen, all work proceeds uniformly within it.

What follows. Part V specifies the role of VR numbers in VR-Sets: the von Neumann ordinals O_n and ω itself are specific examples of well-founded sets, landing in the ZFC-mode as a theorem. Arithmetic is inherited via direct isomorphism with VR Part I (no separate proofs are required). Part V is a short part fixing the correspondence and establishing that the numbers of VR in VR-Sets are precisely the numbers of classical arithmetic, presented operationally.